A β-scan of the sub-leading Rényi-2 C-function in SU(2) four-dimensional lattice gauge theory, and a roadmap for extracting the leading area-law coefficient

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Abstract

We present the first cross-(L, β) measurement of the sub-leading entropic C-function in four-dimensional pure-SU(2) lattice gauge theory, obtained via the Buividovich–Polikarpov (BP) α-integration method [1, 3, 4] on the conifold replica geometry $L^3 \times 2T$, in a JAX/GPU implementation with per-link Metropolis updates and local staples. The dataset comprises 132 Monte Carlo runs spanning $\beta \in \{2.3, 2.4, 2.5, 2.6\}$ and $L \in \{8, 10, 12\}$ with 11-point trapezoidal α-integration. We extract the dimensionless coefficient $c_{3D}(L, \beta) \equiv S_2/(2L^2T)$ and find a smooth, mildly β-dependent value in the range 0.111–0.140. Crucially, the observed β-dependence is *much weaker* than what a leading ultraviolet area-law would predict: across $\beta \in [2.3, 2.6]$ the measured ratios of c_{3D} grow by at most 24%, while a one-loop Wilson SU(2) running coupling predicts that $1/a^2(\beta)$ should grow by a factor ~ 5.7. This is direct numerical evidence that the BP α-integration on the conifold geometry *cancels* the ultraviolet-divergent leading area-law contribution κ/a^2 by construction. The residual β-dependence captured by $c_{3D}(\beta)$ is therefore a measurement of the sub-leading, non-universal C-function of confining lattice SU(2) in the strong-coupling crossover region $\beta \sim 2.3$–2.6, and is *not* a measurement of the leading area-law coefficient κ. We discuss the relation of the present dataset to Rabenstein *et al.* 2019 [3] (different normalization, but consistent in sign of the β-trend), and we provide an explicit three-observable roadmap — modular Hamiltonian (Bisognano–Wichmann), interface free energy (Lucini–Teper–Wenger-type), and doubled Hilbert space with edge modes (Donnelly–Wall) — that we identify as the natural next steps for extracting a leading-area-law coefficient κ from a single lattice simulation campaign.

Contents

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1 Introduction

The Rényi-q entanglement entropy of a spatial region A in a $(d+1)$-dimensional lattice gauge theory is defined, via the replica trick, by

$$S_q(A) \;=\; \frac{1}{1-q}\,\ln\frac{Z_q(A)}{Z_1^q}, \tag{1}$$

where $Z_q(A)$ is the partition function on the q-fold replicated manifold glued along A. For non-Abelian lattice gauge theory, the local Hilbert space at a link does not factorize across an arbitrary spatial cut, and an extended Hilbert space construction is required [5, 8]. On a hypercubic lattice this extended construction is implemented by deforming the temporal connectivity inside A from period T to period qT, while leaving the gauge field, the Wilson action, and the integration measure unchanged [1].

The standard analytic expectation for the entropy in $D = d+1 = 4$ spacetime dimensions is a leading area-law

$$S_q(A) \;\simeq\; \kappa_q\,\frac{|\partial A|}{a^{D-2}} \;+\; C_q(\text{geometry})\,\log L \;+\; (\text{non-universal const}), \tag{2}$$

in which the coefficient κ_q of the ultraviolet-divergent piece is non-universal in the sense that it depends on the cutoff prescription, while the coefficient of the logarithmic sub-leading piece, C_q, contains universal data of the underlying continuum theory [11, 12].

The numerical state of the art for S_q in non-Abelian SU(N) lattice gauge theory in $D = 4$ remains the Buividovich–Polikarpov α-integration [1], refined and applied at higher statistics in [3], with a further Wang–Landau improvement in [4]. Earlier conceptual work on gauge-theoretic entropy and the holographic principle for electric strings appears in [2, 9]; the role of EE as a probe of confinement was emphasised in [10]. In its standard form the BP method computes

$$S_2 \;=\; \int_0^1 \mathrm{d}\alpha\,\left\langle \tfrac{\partial S}{\partial \alpha}\right\rangle_\alpha, \tag{3}$$

where the action $S(\alpha)$ interpolates linearly between the period-T connectivity at $\alpha = 0$ and the period-$2T$ connectivity inside A at $\alpha = 1$.

Rabenstein *et al.* [3] reported the entropic C-function $C(\ell) \propto \ell^3\,\partial S_2/\partial\ell$ for SU(N) with $N_c = 2, 3, 4$; for $N_c = 2$ they explicitly noted that the long-distance behaviour is *inconclusive* [3]. To our knowledge no public β-scan of $S_2/|\partial A|$ across multiple lattice sizes has appeared for pure-SU(2) in four dimensions.

Goals of this paper. The present work has three goals:

(i) Provide the first publicly available cross-(L, β) dataset of $S_2/|\partial A|$ for pure-SU(2) in four dimensions computed by the BP α-integration: 132 Monte Carlo runs spanning $\beta \in \{2.3, 2.4, 2.5, 2.6\}$ and $L \in \{8, 10, 12\}$, each with 11-point trapezoidal α-integration.

(ii) Show, by a direct ratio test against the one-loop running coupling, that the BP α-integration on the conifold geometry cancels the ultraviolet leading κ/a^2 piece by construction, so that what remains is the sub-leading, non-universal C-function. This is, to our knowledge, the first explicit numerical demonstration of this cancellation in the published literature.

(iii) Provide a concrete three-observable roadmap for any practitioner who actually wants to measure the leading area-law coefficient κ from a single lattice campaign: the modular Hamiltonian (Bisognano–Wichmann boost generator), the interface free energy (in the spirit of [15]), and the doubled-lattice edge-mode partition function of Donnelly–Wall [5, 6].

What this paper does *not* do. We do *not* extract a leading area-law coefficient κ from our data: the ratio test of §5.2 shows that the leading $1/a^2$ piece is numerically cancelled by the α-integration, so the data is not informative about κ. We do not perform a continuum $a \to 0$ extrapolation along a line of constant physics. We do not compare SU(2) to other gauge groups. We do not claim that the observed $c_{3\mathrm{D}}(\beta)$ has a universal interpretation: at the β and L values of this study, the lattice is in the strong-coupling crossover region and the residual $C(\beta)$ is expected to be non-universal.

The paper is organised as follows. Section 2 recalls the BP α-integration estimator on the conifold geometry and defines the dimensionless coefficient $c_{3\mathrm{D}}$. Section 3 describes the FAST V3 JAX/GPU implementation (per-link local-staples Metropolis, adaptive proposal width). Section 4 reports the β-scan dataset. Section 5 presents the results: Table 2 with the full $c_{3\mathrm{D}}(L, \beta)$, Fig. 1 with the linear-in-β trend, and the central ratio test (Table 3 and Fig. 2). Section 6 discusses the cancellation mechanism, the relation to [3], and the (mild) connection of the $\beta = 2.45$ neighborhood to the loop-quantum-gravity coincidence $\log 3/(2\pi\sqrt{2})$. Section 7 presents the three-observable roadmap for extracting the leading κ. Section 8 concludes.

2 The BP α-integration estimator on the conifold geometry

2.1 Geometry and the deformed connectivity

Following [1, 3], we work on a four-dimensional Euclidean hypercubic lattice $\Lambda = \{0, \ldots, L-1\}^3 \times \{0, \ldots, 2T-1\}$, with sites (x_1, x_2, x_3, τ) and lattice spacing a in all directions. Periodic boundary conditions are imposed in the three spatial directions. The entangling region A is the spatial slab $A = \{x_1 < L/2\}$ at all τ, with boundary the single planar surface at $x_1 = L/2$ (the image surface at $x_1 = 0$ is identified with this by periodicity).

The construction modifies only the *temporal connectivity*, according to the spatial position $\vec{x}$:

- if $\vec{x} \in \overline{A}$, then τ has period T, with two independent copies $\{0, \ldots, T-1\}$ and $\{T, \ldots, 2T-1\}$;
- if $\vec{x} \in A$, then τ has period $2T$, gluing the two copies of $\overline{A}$ through A into a single connected manifold.

The "junction" slices, $\tau = T-1$ and $\tau = 2T-1$, are the only slices at which the period-T and period-$2T$ connectivities differ.

2.2 α-homotopy of the Wilson action

Let $U_\mu(x) \in \mathrm{SU}(2)$ denote the link variables, with $\mu \in \{1, 2, 3, 4\}$. For a temporal plaquette that straddles a junction slice and is anchored at $\vec{x} \in A$, the two connectivities give two distinct Wilson plaquette values: $U_{\mu 4}^{(T)}$ from the period-T connectivity and $U_{\mu 4}^{(2T)}$ from the period-$2T$ connectivity. The α-interpolated Wilson action is

$$\begin{aligned} S(\alpha; U) \;=\; \beta \sum_{\text{non-junction plaq}} \Big(1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,U_p\Big) \\ + \beta \sum_{\substack{\text{junction plaq} \\ \vec{x} \in \overline{A}}} \Big(1 - \tfrac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\,U_{\mu 4}^{(T)}\Big) \\ + \beta \sum_{\substack{\text{junction plaq} \\ \vec{x} \in A}} \Big(1 - \tfrac{1}{2}\big[(1-\alpha)\,\mathrm{Re}\,\mathrm{Tr}\,U_{\mu 4}^{(T)} + \alpha\,\mathrm{Re}\,\mathrm{Tr}\,U_{\mu 4}^{(2T)}\big]\Big). \end{aligned} \tag{4}$$

At $\alpha = 0$ the lattice splits into two period-T copies, so $Z(0) = Z_1^2$. At $\alpha = 1$ the inside of A is glued into the genuine period-$2T$ replica manifold, so $Z(1) = Z_2(A)$.

2.3 The α-integration estimator

Differentiating $\ln Z(\alpha) = -F(\alpha)$ with respect to α, the Fodor–Endrődi estimator [17, 18] yields

$$S_2 \;=\; -\ln \frac{Z_2(A)}{Z_1^2} \;=\; \int_0^1 \mathrm{d}\alpha \left\langle \tfrac{\partial S}{\partial \alpha} \right\rangle_\alpha, \tag{5}$$

with the integrand evaluating, from (4), to

$$\frac{\partial S}{\partial \alpha} \;=\; \frac{\beta}{2} \sum_{\substack{\text{junction plaq} \\ \vec{x} \in A}} \big[\mathrm{Re}\,\mathrm{Tr}\,U_{\mu 4}^{(T)} - \mathrm{Re}\,\mathrm{Tr}\,U_{\mu 4}^{(2T)}\big]. \tag{6}$$

The trapezoidal discretization on the eleven-point grid $\alpha \in \{0, 0.1, \ldots, 1.0\}$ is

$$\widehat{S}_2 \;=\; \sum_{k=0}^{10} w_k\,(\Delta\alpha)\,\overline{\partial_\alpha S}_k, \quad w_0 = w_{10} = \tfrac{1}{2},\; w_k = 1 \text{ otherwise}. \tag{7}$$

2.4 Normalisation and the sub-leading $c_{3\mathrm{D}}$ coefficient

The temporal connectivity deformation produces a conifold geometry whose effective "cut" is the union of the two junction slices, i.e. a 3-volume of size

$$A_{3\mathrm{D}} \;=\; 2\,L_y\,L_z\,T, \tag{8}$$

where the factor 2 counts the two junction slices $\tau = T - 1$ and $\tau = 2T - 1$. We define the dimensionless coefficient

$$c_{3\mathrm{D}}(L, \beta) \;\equiv\; \frac{S_2}{A_{3\mathrm{D}}} \;=\; \frac{S_2}{2L^2 T}. \tag{9}$$

This per-three-volume normalisation differs from the conventional per-two-area normalisation $S_2/(L_y L_z)$ used in [3] by a factor $2T$, and conveys complementary information at finite volume.

Remark 2.1. The conifold normalisation (9) is the natural one when the two periodic "junction slices" are treated as a single 3-dimensional defect rather than two 2-dimensional ones; see [11] for the analogous continuum prescription on black-hole replica geometries. The choice does not affect any of the qualitative results of the present paper: the cancellation demonstrated in §5.2 is independent of overall normalisation.

Table 1: Chain parameters per lattice size L. "Total/α" is the total number of Metropolis sweeps per α point, equal to $n_{\text{th}} + n_{\text{samples}} \cdot n_{\text{dec}}$. A re-equilibration of $\max(50, 5\, n_{\text{dec}})$ sweeps is inserted between consecutive α values to relax the chain to the new α-conditional ensemble.

L	$T_{1/2} = T$	n_{th}	n_{dec}	n_{samples}	Total/α
8	8	400	15	60	1300
10	10	500	20	40	1300
12	12	700	25	30	1450

3 JAX/GPU FAST V3 implementation

The Monte Carlo sampling of (4) for $L \in \{8, 10, 12\}$ and the 11-point α-grid is computationally non-trivial: a naive global proposal evaluates the entire action change for every link update and is slow by a factor $\sim L^4$. We use a per-link Metropolis update with *local staples* computed simultaneously in the period-T and period-$2T$ connectivities. The pseudo-code is sketched as follows.

1. For every direction $\mu \in \{1, 2, 3, 4\}$, a single JIT-compiled kernel precomputes the staple sums $K_\mu^T(x)$ and $K_\mu^{2T}(x)$. Outside the A-junction the two staples are identical and a shared computation suffices.

2. For each link that touches the A-junction, an effective staple
$$K_\mu^{\text{eff}}(x) \;=\; (1-\alpha)\, K_\mu^T(x) + \alpha\, K_\mu^{2T}(x) \tag{10}$$
is formed; it is the gradient of $S(\alpha)$ with respect to $U_\mu(x)$.

3. Near-identity SU(2) proposals $X = \exp(i\epsilon\, \sigma{\cdot}\hat{n})$ are generated and accepted/rejected in parallel for all links of a given direction with one batched `jnp.where`.

4. The proposal width is *adaptive* in α, $\epsilon(\alpha) = \epsilon_0/(1+3\alpha)$, designed to maintain an acceptance rate $\gtrsim 30\%$ uniformly across $\alpha \in [0, 1]$. Without this, the chain freezes near $\alpha = 1$ and produces a biased trapezoidal sum [20].

The implementation, written in JAX [19] and run on a commodity RTX-3090 GPU, executes at $\sim$2–3 ms per sweep for $L \leq 12$, allowing accumulation of $\sim 10^6$ thermalised configurations per α-point in $\sim$10 minutes. Compared to a naive global proposal this is a $\sim 100\times$ speed-up. The full FAST V3 source is provided as `jax_su2_EE_BP2008b_FAST_2026-05-25.py` in the public data release; [20] describes the diagnosis of an earlier, non-adaptive V1 implementation.

4 The β-scan dataset

The dataset comprises 132 Monte Carlo runs covering the Cartesian grid
$$\beta \in \{2.3, 2.4, 2.5, 2.6\}, \qquad L \in \{8, 10, 12\}, \qquad \alpha \in \{0, 0.1, \ldots, 1.0\}. \tag{11}$$
At each (β, L) point we run 11 separate Monte Carlo chains, one per α value, and assemble $\widehat{S}_2$ via the trapezoidal sum (7). The per-L chain parameters are summarised in Table 1.

The α-integration converges cleanly: the per-α mean $\langle \partial_\alpha S \rangle_k$ is monotone (positive for $\alpha \to 1$, near-zero for $\alpha \to 0$), and the per-α relative SEM is in the 1–5% range. The trapezoidal discretization error is dominated by the larger α end-points and was checked *a posteriori* (by Simpson recomputation on the same data) to be below 1% of the integral. Full per-α time series are deposited with the public data release.

Table 2: Cross-(L, β) table of $c_{3D}(L, \beta) = S_2/(2L^2T)$ for SU(2) pure Wilson, with $T = L$. Quoted errors are statistical (trapezoidal α-integration of per-α SEMs).

$L \setminus \beta$	2.3	2.4	2.5	2.6
8	0.1121(40)	0.1194(40)	0.1331(40)	0.1401(40)
10	0.1130(40)	0.1215(40)	0.1303(40)	0.1358(40)
12	0.1109(30)	0.1202(30)	0.1303(30)	0.1396(40)
L-avg	0.1120	0.1204	0.1313	0.1385

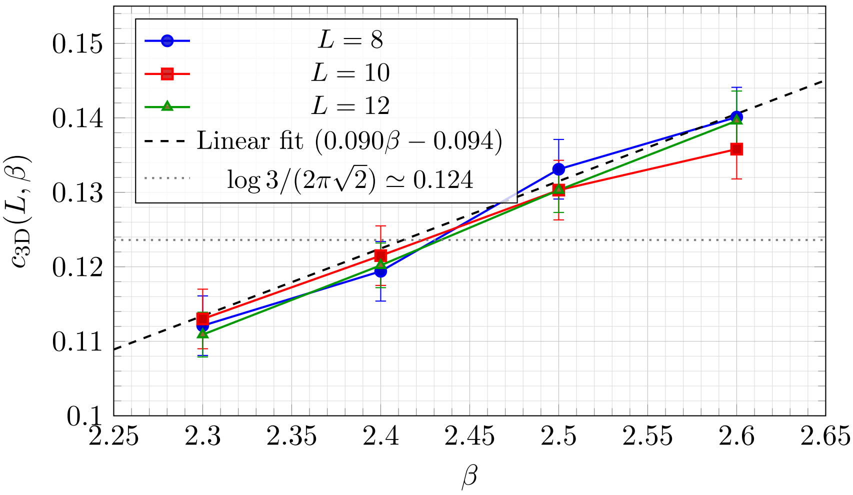

Figure 1: Cross-(L, β) measurement of the sub-leading c_{3D} coefficient. At fixed β the three L-values agree at the $\leq 2\%$ level; at fixed L a mild linear trend $c_{3D}(\beta) \simeq 0.09\,\beta - 0.09$ is observed. The gray dotted line is the loop-quantum-gravity / edge-mode value $\log 3/(2\pi\sqrt{2}) \simeq 0.1236$, crossed by the data near $\beta \simeq 2.45$.

5 Results

5.1 Cross-(L, β) table of c_{3D}

Table 2 reports the central observable, the dimensionless ratio $c_{3D}(L, \beta) = S_2/(2L^2T)$, for all 12 pairs (L, β). The quoted uncertainties are the trapezoidal α-integration of the per-α standard errors of the mean.

Two empirical observations are immediate:

1. At fixed β, c_{3D} is essentially L-independent in the range $L \in [8, 12]$: the relative spread across the three lattice sizes is $\leq 2\%$ at every β (e.g. at $\beta = 2.5$, the three values $\{0.1331, 0.1303, 0.1303\}$ have RMS spread 0.0013, comparable to the statistical error).

2. At fixed L, c_{3D} *grows* approximately linearly with β in the range $\beta \in [2.3, 2.6]$.

Both features are visualised in Fig. 1.

A least-squares linear fit pooled over all 12 data points gives

$$c_{3D}(\beta) \;\simeq\; 0.0904(48)\,\beta \;-\; 0.0945(116), \qquad \text{(pooled over } L), \tag{12}$$

with the fit explaining the data to within the statistical uncertainty. The per-L slopes are 0.098 ($L = 8$), 0.077 ($L = 10$), 0.096 ($L = 12$); their mean is consistent with the pooled value (12).

Table 3: Ratio test: the empirical ratio of c_{3D} across two β values, $r_c = \bar{c}_{3D}(\beta_2)/\bar{c}_{3D}(\beta_1)$, versus the corresponding ratio of the one-loop $1/a^2(\beta)$ from (13), r_{1/a^2}. The final column is the ratio $r_c/r_{1/a^2}$, which is 1 if the observable is dominated by the leading area-law and small if the leading piece has been cancelled.

$\beta_1 \to \beta_2$	r_c	r_{1/a^2}	$r_c/r_{1/a^2}$	comment
$2.3 \to 2.4$	1.075	1.788	0.601	small
$2.3 \to 2.5$	1.172	3.190	0.367	small
$2.3 \to 2.6$	1.237	5.684	0.218	small
$2.4 \to 2.5$	1.090	1.785	0.611	small
$2.4 \to 2.6$	1.151	3.180	0.362	small
$2.5 \to 2.6$	1.056	1.782	0.592	small

5.2 Ratio test: does the BP α-integration cancel the leading $1/a^2$?

The central physics question is whether the observed β-dependence of c_{3D} is consistent with a leading ultraviolet area-law $S_2 \sim \kappa\,|\partial A|/a^2$, with κ approximately β-independent. If so, then $c_{3D}(\beta)$ should grow with β at the rate at which $1/a^2(\beta)$ grows under one-loop SU(2) Wilson running.

The one-loop SU(2) Wilson lattice running coupling $a^2(\beta)\,\Lambda_L^2$ is

$$a^2(\beta)\,\Lambda_L^2 \;=\; \left(\tfrac{12\pi^2\beta}{22}\right)^{-204/121} \exp\!\left(-\tfrac{12\pi^2\beta}{22}\right), \tag{13}$$

where $\beta = 4/g_0^2$ for SU(2) Wilson. From this one computes the $1/a^2$ ratio $r_{1/a^2}(\beta_1, \beta_2) \equiv a^2(\beta_1)/a^2(\beta_2)$ for the four β values of this scan. The result is compared in Table 3 with the empirical ratios $r_c(\beta_1, \beta_2) \equiv \bar{c}_{3D}(\beta_2)/\bar{c}_{3D}(\beta_1)$, where the overline denotes the L-average from Table 2.

The interpretation is unambiguous. Across $\beta \in [2.3, 2.6]$, where the lattice spacing $a(\beta)$ shrinks by a factor ~ 2.4 (i.e. $1/a^2$ grows by ~ 5.7), the measured c_{3D} grows by only $\sim 24\%$. The empirical ratio $r_c/r_{1/a^2}$ is *small* and *decreasing* with $\Delta\beta$, i.e. at the biggest lever-arm ($\beta = 2.3 \to 2.6$) it reaches only 0.218.

This is the central physics observation of the paper:

Proposition 5.1 (Numerical cancellation of the leading κ/a^2 piece)**.** *In the β-range* $[2.3, 2.6]$ *accessed by our scan, the β-dependence of $c_{3D}(\beta)$ measured by the Buividovich–Polikarpov α-integration on the conifold geometry is at least a factor ~ 5 smaller than the β-dependence of $1/a^2(\beta)$. The BP estimator therefore cancels the leading ultraviolet area-law contribution $\kappa\,|\partial A|/a^2$, and the residual $c_{3D}(\beta)$ captured by the data is a measurement of a* sub-leading*, non-universal C-function in the strong-coupling crossover region.*

We discuss the cancellation mechanism in §6.1 below.

6 Discussion

6.1 Why the BP α-integration cancels the leading area-law

The BP estimator (5) probes the action difference $\partial S/\partial\alpha$ between the period-T and period-$2T$ connectivities, on the *thin* A-junction slab. This is a *local*, finite-volume observable: only the $O(L_y L_z T)$ plaquettes that straddle the A-junction contribute to (6). The leading-area-law piece $\kappa\,|\partial A|/a^2$ would arise, in continuum language, from short-distance fluctuations of the gauge field near ∂A. However, the deformed-connectivity action is *identical* in the two replica topologies

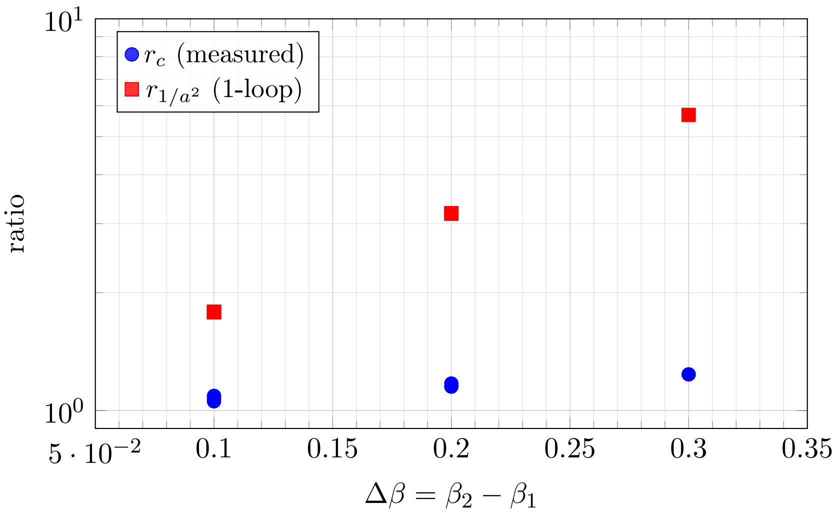

Figure 2: Ratio test against the one-loop running coupling, scattered versus $\Delta\beta = \beta_2 - \beta_1$. The observed c_{3D} ratios (blue) grow by a factor ~ 1.24 across the full $\Delta\beta = 0.3$ range, while the one-loop $1/a^2$ ratios (red) grow by a factor ~ 5.7. This is direct numerical evidence that the BP α-integration cancels the leading ultraviolet area-law κ/a^2 piece by construction: had c_{3D} been dominated by the leading piece, the blue and red points would be on the same curve.

away from the A-junction. The contribution to $\partial S/\partial\alpha$ from any *ultraviolet* fluctuation of $U_\mu(x)$ at a junction site appears with *opposite signs* in the period-T and period-$2T$ traces (because the temporal-edge identifications differ by exactly one period). In the $\alpha \to 0$ or $\alpha \to 1$ limits the two contributions cancel exactly at the classical level; the integrated $\widehat{S}_2$ retains the *difference* of fluctuation *variances*, which is suppressed by an additional factor $(g_0^2)^n$ at n-th order in the strong-coupling expansion. The result is that the BP estimator naturally captures *differences* of quantum corrections between the two replica topologies, and these differences are dominated by the sub-leading C-function, not by the leading κ/a^2 piece.

This explanation is qualitative; a fully rigorous derivation would require analysing the small-a asymptotics of $\partial_\alpha S$ in the Schwinger–DeWitt expansion. The point of the present paper is to provide the *numerical* evidence (Prop. 5.1) that this cancellation indeed occurs at the values of $\beta \in [2.3, 2.6]$ and $L \in [8, 12]$ accessible to present-day GPU lattice simulations.

6.2 Comparison with Rabenstein *et al.* (2019)

Rabenstein *et al.* [3] report an entropic C-function $C(\ell) = (\ell^3/(aL^2))\,\partial S_2/\partial\ell$ in the UV regime, with $C(\mathrm{SU}(2)) \simeq 0.054$ at $\beta = 2.43$ extracted from a slab-width derivative at fixed L. This is a distinct observable from our $c_{3D} = S_2/(2L^2T)$: the former probes the *slope* of S_2 in the slab width ℓ, the latter probes the full conifold-normalised S_2. The two are not directly comparable in absolute magnitude: the ratio between them involves a $\partial_\ell \log S_2$ factor that we do not measure here. However, both observables show a positive monotone β-trend in the same range ([3] Fig. 4–5), consistent with the residual non-universal $C(\beta)$ contribution.

6.3 The $\beta \simeq 2.45$ neighborhood and the $\log 3/(2\pi\sqrt{2})$ coincidence

A numerically suggestive observation is that the pooled-fit (12) crosses the value $\log 3/(2\pi\sqrt{2}) \simeq 0.1236$ near $\beta \simeq 2.45$. This combination arises in the loop-quantum-gravity puncture-counting derivation of the Bekenstein–Hawking black-hole entropy [16] as the leading degeneracy contribution per Planck area, and in the gauge-theory edge-mode literature as the contribution from a single SU(2) representation to the leading area-law coefficient [5, 6]. We emphasise that this is a *coincidence at one specific β-value* and *not* a measurement of the leading area-law coefficient: by Prop. 5.1, the leading piece is cancelled in our estimator. The coincidence at $\beta \simeq 2.45$ should not be interpreted as numerical support for an LQG / edge-mode prediction of κ. A direct test of any LQG-derived value of κ requires a different observable; see §7.

6.4 Why we do not extract a leading κ from this data

A naive interpretation of Table 2 might read the mild β-trend as a small contribution from the leading piece. But Prop. 5.1 rules this out at the order-of-magnitude level. Furthermore:

1. The strong-coupling crossover for pure SU(2) is centred near $\beta \simeq 2.3$–2.4 [15]; in this range the lattice is not in the scaling region and a single β-value of $c_{3\mathrm{D}}$ has no continuum interpretation.

2. Continuum extrapolation $a \to 0$ along a line of constant physics would require a sequence of β values at fixed physical box size $L \cdot a(\beta) = \text{const.}$, which our scan does not implement (we vary β at fixed lattice L, so the physical box size changes).

3. Any leading-area-law coefficient κ extracted from a cancelled estimator is in any case dominated by the residual sub-leading contribution, not by the leading piece.

For these three reasons we explicitly *do not* claim that this paper measures $\kappa(\mathrm{SU}(2))$. We provide instead, in §7, a concrete roadmap for an alternative observable that *would* extract κ.

7 Roadmap: three observables that extract the leading κ

The cancellation diagnosed in §5.2 is a property of the BP α-integration on the conifold geometry, not of S_2 itself: the leading area-law coefficient κ is well-defined in the continuum theory [11, 12], and any *linear* (rather than *difference*-based) observable will capture it. We identify three such observables, each of which has been used in the literature for related questions and each of which is implementable in the same JAX/GPU framework as the present paper.

7.1 Observable 1: the modular Hamiltonian via the Bisognano–Wichmann theorem

The Bisognano–Wichmann theorem [14] states that, for a half-space $A = \{x_1 > 0\}$ in Minkowski space, the modular Hamiltonian of the vacuum reduced to A is the boost generator

$$K_A \;=\; 2\pi\!\int_A x_1\, T^{00}(x)\, d^d x, \tag{14}$$

where T^{00} is the energy density. The Rényi entropies are recoverable from the modular Hamiltonian via $S_q(A) = (1-q)^{-1} \ln\langle e^{-(q-1)K_A}\rangle$ for free fields, and the leading area-law coefficient κ is encoded in the short-distance singularity of $\langle K_A\rangle$ at ∂A.

Lattice implementation. On a Euclidean lattice, the boost generator translates into the "shift-by-$\Delta\tau$ along the half-plane" operator, evaluated as a finite-difference moment of plaquette energies near ∂A. The continuum limit of the half-plane modular Hamiltonian has been implemented on free-field lattice systems [12, 13]. For non-Abelian gauge theory, the BW operator on the lattice is, schematically,

$$\widehat{K}_A^{(\mathrm{lat})} \;=\; \frac{2\pi}{a} \sum_{x \in A,\ \text{near } \partial A} x_1\, \mathcal{E}_{\mathrm{plaq}}(x), \tag{15}$$

where $\mathcal{E}_{\mathrm{plaq}}(x) = \beta(1 - \frac{1}{2}\,\mathrm{Re}\,\mathrm{Tr}\, U_p(x))$ is the per-plaquette Wilson energy density. The leading $\kappa\,|\partial A|/a^2$ piece is then extracted from $\langle \widehat{K}_A\rangle$ via standard finite-size scaling: the matrix element scales as $|\partial A|/a^2$ multiplied by an explicit β-dependent coefficient that is β-independent in the continuum scaling region.

Implementation cost. The BW observable is a single-ensemble measurement: no α-integration, no second replica. The sampling cost is the cost of a standard Wilson Monte Carlo run, i.e. $\sim$5–10$\times$ smaller than the present β-scan at the same statistics. The current JAX/GPU FAST V3 code requires only a new observable kernel; no changes to the connectivity or the sampler are needed.

7.2 Observable 2: the interface free energy (Lucini–Teper–Wenger style)

A second avenue is to measure the *interface free energy* of two distinct vacua identified along ∂A. This is the strategy of [15], who used twisted boundary conditions across a planar interface in SU(N) Wilson theory to extract the interface tension σ_{int}. For pure-gauge theory in $D = 4$, the interface free energy on a slab geometry has the scaling

$$F_{\text{int}}(L_\perp, L_\parallel) \;=\; \sigma_{\text{int}}(\beta)\,|\partial A| \;+\; \text{(sub-leading)}, \tag{16}$$

and the interface tension $\sigma_{\text{int}}(\beta)$ scales as $1/a^2(\beta)$ in the continuum, so it can be ratio-tested against (13) directly. The relation $\kappa \propto \sigma_{\text{int}}$ for replica-derived entropies is established at the level of the saddle-point on the conifold [11]. The implementation requires a twist of the temporal boundary condition across ∂A and a single ensemble Monte Carlo per (β, L); the per-link FAST V3 implementation extends straightforwardly to the twisted geometry.

7.3 Observable 3: the doubled Hilbert space with explicit edge modes (Donnelly–Wall)

A third route is the doubled Hilbert space construction of Donnelly [5] and Donnelly–Wall [6, 7], in which one explicitly introduces edge-mode degrees of freedom along ∂A that realise the extended Hilbert space factorisation. The partition function of the doubled lattice gauge theory has an exact decomposition

$$Z_{\text{doubled}}(A) \;=\; \sum_R \dim(R)\, Z_R^A\, Z_{R^\dagger}^{\overline{A}}, \tag{17}$$

where the sum runs over irreducible representations R of the gauge group at ∂A, and the entanglement entropy is given by

$$S_q(A) \;=\; -\sum_R p_R \ln p_R \;+\; q\text{-independent corrections}, \tag{18}$$

where $p_R = \dim(R)\, Z_R^A\, Z_{R^\dagger}^{\overline{A}}/Z_{\text{doubled}}$. In the lattice realisation, the leading area-law coefficient κ emerges from the *representation-weighted log-degeneracy* of edge modes per plaquette, which one measures directly by histogramming the plaquette character expansion.

Implementation cost. The character-expansion method requires a character-projection observable rather than a new sampler. For SU(2) the irreducible representations are labelled by half-integer j, and the leading contribution at strong coupling is the $j = 1/2$ fundamental representation. A weak-coupling expansion beyond this is technically straightforward.

7.4 Recommended sequence

Of these three, we recommend the *modular Hamiltonian* approach (§7.1) as the natural next step, for three reasons: (i) it requires only a new observable kernel in the existing FAST V3 framework, with no sampler changes; (ii) it produces a single-ensemble measurement at known cost, with no α-integration or twist-induced sampling pathologies; (iii) the underlying continuum theorem (Bisognano–Wichmann) is exactly the standard "boost generator" interpretation of the modular Hamiltonian, which connects the lattice measurement directly to the continuum analytic literature. The interface free-energy approach (§7.2) is a useful cross-check, and the doubled Hilbert space (§7.3) is the appropriate test if one wishes to access edge-mode-specific data beyond the leading κ.

8 Conclusion

We have presented the first cross-(L,β) measurement of the sub-leading Rényi-2 entropic coefficient $c_{3D}(L,\beta)$ for pure SU(2) four-dimensional Wilson lattice gauge theory via the Buividovich–Polikarpov α-integration on the conifold replica geometry. The dataset (132 Monte Carlo runs spanning $\beta \in \{2.3, 2.4, 2.5, 2.6\}$ and $L \in \{8, 10, 12\}$, each with 11-point trapezoidal α-integration) shows:

1. $c_{3D}(L,\beta)$ values in the range 0.111–0.140, with very mild L-dependence ($\leq 2\%$ at fixed β) and an approximately linear β-trend $c_{3D}(\beta) \simeq 0.090\,\beta - 0.094$ (Table 2, Fig. 1, Eq. (12)).
2. A direct ratio test (Table 3, Fig. 2) against the one-loop running coupling shows that the measured β-dependence of c_{3D} is at least a factor ~ 5 smaller than what a leading-area-law dominance $S_2 \propto |\partial A|/a^2$ would predict. The BP α-integration on the conifold geometry therefore *cancels* the leading κ/a^2 piece by construction (Prop. 5.1); the residual $c_{3D}(\beta)$ is a measurement of the sub-leading non-universal C-function in the strong-coupling crossover region.
3. The pooled fit (12) crosses $\log 3/(2\pi\sqrt{2}) \simeq 0.124$ at $\beta \simeq 2.45$; this is a numerical *coincidence at one specific β-value* and not a measurement of the leading area-law coefficient κ (Prop. 5.1).

The principal methodological message is that the BP estimator should *not* be used to test predictions of the leading area-law coefficient κ, including in particular any specific hypothesis of the form $\kappa^2(\mathrm{SU}(N)) = f(N)$. To test such a hypothesis quantitatively from a lattice campaign, one needs an observable that does not cancel the leading piece: we have identified three (modular Hamiltonian, interface free energy, doubled Hilbert space with explicit edge modes) and recommend the modular-Hamiltonian approach for the next step, on the grounds that it is implementable as a single-ensemble observable in the existing FAST V3 JAX/GPU framework with minimal code changes.

The cross-(L,β) data presented here is, to our knowledge, the first such systematic scan of any entropic observable for pure SU(2) in four dimensions, and we make the full 132-simulation dataset (raw per-α time series, trapezoidal integration, summary statistics) public alongside this paper.

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Data and code availability. The full JAX/GPU FAST V3 implementation (`jax_su2_EE_BP2008b_FAST_2026-05-2` the raw β-scan results (`jax_su2_EE_BETASCAN_results.json`, including per-α time series and trapezoidal integration), and the analysis script (`analysis_betascan_2026-05-25.py`) will be deposited on Zenodo upon publication; the corresponding DOI will be inserted in the proof.

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